

Sensor Research and Fusion

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Four-sensor roles

Sensor	What it contributes	What it must NOT do
Radar	Range, speed, primary clear-road proof	Cannot reliably name all object classes
RGB	Object naming, road edges, fade-test boards	Cannot clear a radar hazard
Thermal	Warm people/engines/tyres/vehicles	Cannot lengthen proven distance
NIR	Additional dusk/night/dust contrast and reflective markers	Cannot claim visibility through dense fog

2. One-way fusion rule

Radar sets the proven-clear range. If another sensor detects an object nearer than the radar range, the proven distance shortens immediately. If an optical sensor sees nothing, that is not evidence that the road is clear. Sensor failure therefore cannot make the safe corridor larger (p. 4).

Two independent sensor types agreeing on bearing and range can confirm a hazard for warning, but confirmation still cannot clear anything.

3. Fog visibility measurement

ProofRoad proposes roadside fog posts with a scatter/visibility sensor plus temperature and humidity, reporting current visibility and trend rather than a long-range forecast (p. 3).

For camera-based checking, two high-contrast boards at known distances can be used for a fade test. ProofRoad notes an experimental Koschmieder-style relationship and explicitly says calibration is required (p. 9).

The Met Office describes modern visibility sensors as measuring meteorological optical range. This provides useful external context for designing and documenting the fog-post measurement method.

4. Prototype calibration plan

Calibrate each sensor independently before fusion.

Record timestamp/sequence, sensor health, range, bearing and object class/confidence where applicable.

Run paired-sensor tests at controlled distances.

Test disagreement cases deliberately: radar-only obstacle, thermal-only warm object, RGB-only apparent object, and no-sensor-clear.

Never let a failed/blinded sensor increase the corridor.